EgoEMG: A Multimodal Egocentric Dataset with Bilateral EMG and Vision for Hand Pose Estimation

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Abstract

Surface electromyography (sEMG) records muscle activity during hand movement and can be decoded to recover detailed hand articulation. EMG and egocentric vision are complementary for hand sensing: EMG captures fine-grained finger articulation even under occlusion and poor lighting, while vision provides global hand configuration. However, no existing dataset synchronizes both modalities. We present *EgoEmg*, a multimodal egocentric dataset for bimanual hand pose estimation. *EgoEmg* includes bilateral wristband EMG with 16 total channels (8 per wrist) sampled at 2 kHz, 120 Hz IMU, egocentric wide-angle RGB video, external RGB-D video, and mocap-derived hand motion with wrist articulation angles. The dataset covers 41 participants performing 60 gesture classes, including 30 single-hand gestures and 30 bimanual gestures, totaling more than 10 hours of recording. We also introduce a benchmark with three tasks—EMG-to-pose, vision-to-pose, and EMG+vision fusion—under a shared joint-angle prediction target and common generalization split axes (cross-gesture, cross-user, and combined). As baselines, we evaluate EMGFormer for EMG-to-pose, generic ResNet/ViT backbones for vision-to-pose, and a hand-specialized WiLoR baseline. We further study a residual fusion architecture that improves over matched lightweight vision-only baselines while remaining weaker than the strongest specialized vision-only model. Together, *EgoEmg* and its benchmark establish a foundation for future research on multimodal hand pose estimation with EMG and vision.

1 Introduction

Hand pose estimation from wearable sensing is a building block for augmented reality, prosthetic control, and physical intelligence [Salter et al., 2024, Atzori and Müller, 2014, Zhao et al., 2026]. Surface EMG is attractive because it measures muscle activation directly and remains usable under self-occlusion, motion blur, and challenging lighting [Farina et al., 2014]. Recent work has shown that EMG can recover detailed hand articulation [Salter et al., 2024, Verma, 2025, Lovecchio et al., 2025]. At the same time, egocentric vision has emerged as a powerful modality for hand tracking in first-person views [Potamias et al., 2024a, Kwon et al., 2021], but relies on line-of-sight and struggles with fast motion blur and self-occlusion during manipulation. These properties suggest a potential complementarity between the two modalities: EMG provides fine-grained finger articulation even when the hand is occluded, while vision recovers global hand configuration and spatial context. Unlocking this complementarity requires a dataset that synchronizes both modalities with pose labels—yet current EMG pose datasets provide no synchronized egocentric video.

We address these gaps with *EgoEmg*, a controlled multimodal egocentric dataset for bimanual hand pose estimation. *EgoEmg* contains synchronized bilateral wristband EMG (16 total channels, 8 per wrist at 2 kHz), per-hand IMU, egocentric wide-angle RGB, external RGB-D, and hand motion

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reconstructed into MANO parameters [Romero et al., 2017] with wrist articulation angles, covering 41 participants and 60 gesture classes (30 single-hand, 30 bimanual) over 10 hours of recording.

We provide a benchmark with three tasks—EMG-to-pose, vision-to-pose, and EMG+vision fusion—sharing the same joint-angle prediction target and generalization splits (cross-gesture, cross-user, and combined). On EMG-to-pose, EMGFormer-S outperforms the prior TDS+LSTM baseline by 22% while using 45% fewer parameters. On vision-to-pose, ResNet backbones outperform generic ViT regressors at lower parameter counts, while the hand-specialized WiLoR baseline achieves the strongest standalone visual accuracy. A residual EMG+vision fusion architecture consistently improves over matched lightweight vision-only baselines, suggesting EMG provides complementary pose information beyond single-frame visual features.

Our contributions are:

- *EgoEmg*, to our knowledge the first dataset jointly providing bilateral wristband EMG and IMU, egocentric RGB, external RGB-D, bimanual MANO annotations with wrist joint angles, and per-frame gesture class labels—released publicly upon acceptance.
- A three-task hand pose estimation benchmark (EMG-to-pose, vision-to-pose, EMG+vision fusion) with cross-gesture, cross-user, and combined generalization splits under a unified joint-angle prediction target.
- Baseline results showing that EMGFormer provides a strong EMG-to-pose reference on both EMG2Pose and *EgoEmg*, that hand-specialized vision modeling further improves over generic ResNet/ViT baselines in this egocentric setting, and that EMG+vision fusion consistently improves over matched lightweight vision-only baselines.

2 Related work

Hand pose datasets and benchmarks. Public EMG datasets cover gesture recognition [Atzori and Müller, 2014, Pradhan et al., 2022, Kaczmarek et al., 2019, Du et al., 2017, Amma et al., 2015], high-density recordings [Jiang et al., 2021], and demographically diverse cohorts [Gowda et al., 2025]. For continuous pose estimation, EMG2Pose [Salter et al., 2024] provides synchronized bilateral EMG with mocap-derived joint angles, while MyoKi [Andreas et al., 2025] and Jarque-Bou et al. [Jarque-Bou et al., 2019] pair forearm EMG with instrumented-glove kinematics. In contrast, visual hand-pose benchmarks provide rich RGB annotations for monocular, egocentric, and hand-object settings [Zimmermann et al., 2019, Yu et al., 2022, Moon et al., 2023, Banerjee et al., 2024, Kwon et al., 2021, Huang et al., 2024], but do not include synchronized wrist EMG. Existing resources therefore lack a unified benchmark that combines egocentric video, bilateral wristband EMG, wrist articulation, and continuous hand-pose labels, limiting the study of multimodal pose regression from both visual observations and muscle activity.

EMG-based pose estimation. Decoding hand kinematics from surface EMG has progressed from single-DoF and multi-DoF finger-angle regression to full-hand pose estimation. EMG2Pose [Salter et al., 2024] established a large-scale EMG-to-pose benchmark using a TDS+LSTM architecture, demonstrating that wrist-worn EMG can recover continuous hand articulation across many users. Recent work further improves reconstruction fidelity through tendon-driven motion modeling [Verma, 2025] and vector-quantized pose tokenization [Lovecchio et al., 2025]. In parallel, EMG foundation models [Kurbis et al., 2025, Fasulo et al., 2025, Mehlman et al., 2025] explore large-scale pretraining, but primarily target gesture classification rather than continuous pose. Despite these advances, EMG-only pose estimation remains sensitive to inter-user variability, and datasets based on instrumented gloves [Andreas et al., 2025, Jarque-Bou et al., 2019] provide limited degrees of freedom and can suffer from sensor drift and hysteresis.

Vision-based hand pose estimation and multimodal fusion. Monocular RGB hand-pose estimation has advanced rapidly with the MANO hand model [Romero et al., 2017], large-scale annotated datasets [Zimmermann et al., 2019, Yu et al., 2022, Moon et al., 2023, Huang et al., 2024], and Transformer-based reconstructors [Potamias et al., 2024a,b]. Egocentric systems and datasets such as UmeTrack [Han et al., 2022], HOT3D [Banerjee et al., 2024], H2O [Kwon et al., 2021], and WristPP [Xi et al., 2026] further support first-person hand tracking and interaction understanding in wearable and egocentric settings. However, vision-based methods remain fundamentally constrained by line of sight: self-occlusion, inter-hand occlusion, object manipulation, and motion blur can

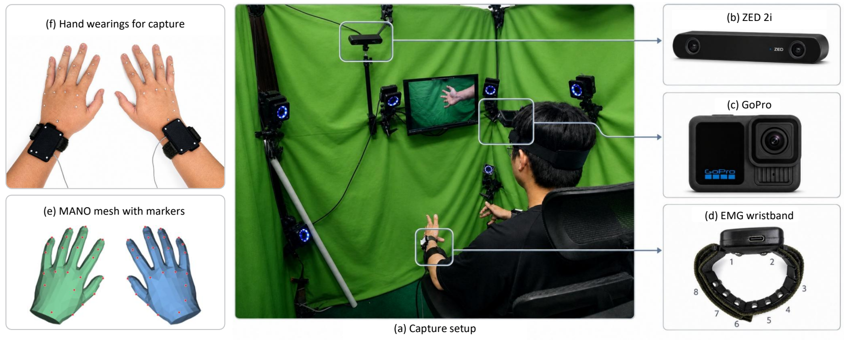

Figure 1: Data collection setup for *EgoEmg*. The capture system consists of bilateral EMG wristbands, a head-mounted egocentric RGB camera, an external ZED 2i RGB-D camera, and an optical motion capture system. Participants wear infrared-reflective markers on both hands, which are used to reconstruct ground-truth hand pose labels.

substantially degrade pose estimates. EMG provides complementary muscle-activation cues that are independent of image visibility, but existing EMG+vision systems have mainly studied gesture classification [Hao et al., 2024, Zandigohar et al., 2024]. The lack of synchronized egocentric video, bilateral EMG, and continuous pose labels has therefore prevented systematic study of EMG+vision fusion for hand-pose regression.

3 *EgoEmg* Dataset

3.1 Overview and Capture Setup

EgoEmg is collected with a synchronized multimodal capture setup, including bilateral EMG, IMU, egocentric RGB, external RGB-D, and optical motion capture. Each participant wears two WAVELETECH EMG wristbands, one on each wrist, with each wristband recording 8 channels of surface EMG at 2 kHz. The visual setup contains a head-mounted wide-angle RGB camera for egocentric hand observation and an external ZED 2i RGB-D camera mounted on the motion-capture rack. Hand motion is tracked using 21 reflective markers per hand at 120 Hz, and all modalities are temporally synchronized.

We recruited 41 participants, aged 18–35 years, with a mean age of 24 years. The cohort includes 23 male and 18 female participants and is balanced across hand dominance. Each participant contributed approximately 15 minutes of recording. *EgoEmg* contains 60 gesture classes, including both single-hand gestures and bimanual interaction gestures. Each frame is annotated with the active gesture class, enabling gesture-conditioned evaluation. During recording, gesture prompts are displayed in randomized order. For single-hand gestures, participants perform the same gesture with both hands in a mirror-symmetric manner while continuously varying wrist orientation. For bimanual gestures, participants follow demonstrated two-hand interactions. This protocol captures both within-gesture wrist variation and natural bimanual coordination patterns.

3.2 Pose Label Reconstruction

Hand pose labels are reconstructed from optical motion-capture markers into MANO parameters [Romero et al., 2017]. We use a learning-based *markers2mano* pipeline that maps 21 marker positions to MANO pose, shape, and translation parameters. The pipeline uses fixed MANO surface vertices as virtual marker correspondences, allowing captured 3D marker positions to be aligned with the canonical MANO mesh. Full details of the *markers2mano* architecture, training data, and augmentation pipeline are provided in Appendix B.1.

Compared with the per-frame inverse-kinematics solver used by EMG2Pose, which has a reported 12.7% invalid-frame rate, our learning-based reconstruction reduces the invalid-frame rate to 3.6%, a 3.5$\times$ reduction. The resulting MANO meshes achieve a mean marker-to-mesh alignment error of

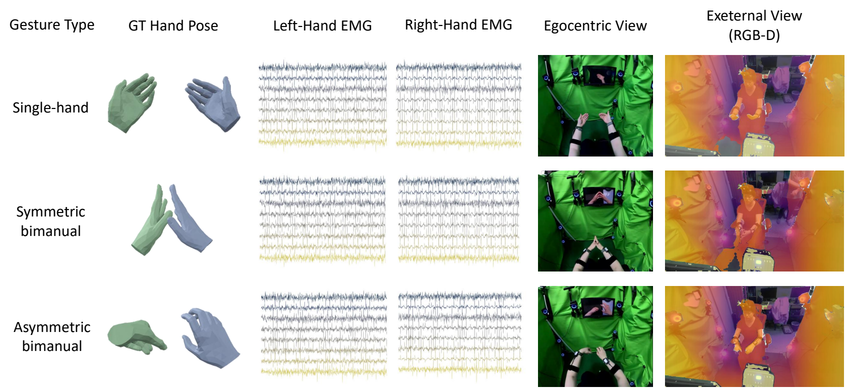

Figure 2: Representative synchronized samples from *EgoEmg*. Each row corresponds to one 3.9 s EMG window. From left to right: ground-truth hand pose, bilateral EMG waveforms, egocentric RGB frame at the window center, and external RGB-D frame.

Table 1: Comparison of *EgoEmg* with representative public EMG datasets. "EMG Ch." counts total recorded channels; "Bilateral" indicates dual-wrist EMG capture; "Wrist labels" indicates explicit wrist articulation labels; "Pose labels" indicates continuous hand pose annotations.

Dataset	Subj.	EMG Ch.	Bilateral	Ego RGB	RGB-D	Wrist labels	Pose labels	Gestures
Jarque-Bou 2019	22	8	–	–	–	–	18 (glove)	26
NinaPro	27–78	10–16	–	–	–	✓†	22 (glove)*	53
MyoKi	36	16	–	–	–	–	18 (glove)	74
PiMForce	21	8	–	–	–	–	20 (glove)	63
EMG2Pose	193	8	✓	–	–	–	20 (mocap)	50
***EgoEmg* (ours)**	**41**	**16**	✓	✓	✓	✓	**22 (mocap)**	**60**

* NinaPro DB1, DB2, and DB5 provide continuous joint angles (22-DoF); remaining sub-databases provide gesture labels only.
(glove) = instrumented glove; (mocap) = optical motion capture. Glove-based angle measurements may be affected by sensor drift, hysteresis, and calibration error.

4.3 mm. We additionally validate reconstructed meshes by projecting randomly sampled meshes onto the egocentric view; details are provided in Appendix B.

Wrist articulation angles are computed directly from motion-capture markers, independent of the MANO reconstruction pipeline. Markers attached to each EMG armband define a forearm coordinate frame, from which we compute two wrist degrees of freedom: flexion/extension and radial/ulnar deviation; the full derivation is provided in Appendix B.2. In addition to MANO parameters, *EgoEmg* provides per-frame joint angles in the 20-DoF scalar format used by EMG2Pose and UmeTrack [Han et al., 2022]. These are obtained by fitting the UmeTrack forward-kinematics mesh to the reconstructed MANO mesh; the conversion procedure is described in Appendix B.3.

3.3 Dataset Statistics and Positioning

Table 1 compares *EgoEmg* with representative public EMG datasets. Among existing datasets, only EMG2Pose and *EgoEmg* provide bilateral EMG with motion-capture-derived continuous hand-pose labels. *EgoEmg* further contributes synchronized egocentric RGB, external RGB-D, IMU, and explicit wrist articulation labels, enabling direct study of cross-modal hand pose estimation.

Hand shape and gesture diversity. *EgoEmg* estimates per-subject MANO shape parameters ($\boldsymbol{\beta} \in \mathbb{R}^{10}$) by averaging shape coefficients for each participant. Figure 3(a) shows the distribution across 41 participants, demonstrating substantial inter-subject hand-shape variation. The 60 gesture classes include 30 single-hand gestures and 30 bimanual interaction gestures, covering both symmetric coordination and asymmetric manipulation. The full gesture vocabulary is provided in Appendix A.1; per-dimension MANO shape statistics are reported in Appendix A.2.

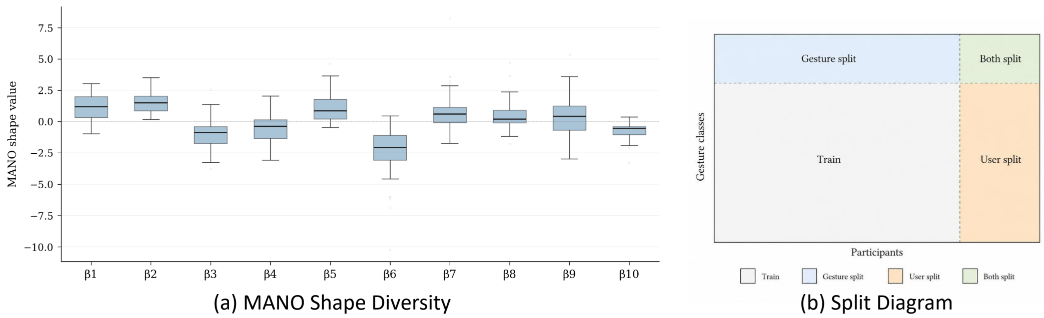

Figure 3: **(a)** MANO shape diversity ($\boldsymbol{\beta} \in \mathbb{R}^{10}$) across *EgoEmg* participants (82 hands). **(b)** Evaluation splits defined on the participant–gesture matrix: Train, Gesture split (blue), User split (orange), and Both split (green).

3.4 Evaluation Splits and Release Format

Following the protocol of EMG2Pose [Salter et al., 2024], we define two primary generalization axes: *gesture* and *user*. Based on these axes, we evaluate three splits: a *gesture split*, where held-out gestures are evaluated on seen participants; a *user split*, where held-out participants are evaluated on all gestures; and a combined *both split*, where both gestures and participants are held out. All modalities follow the same split assignment, preserving synchronized multimodal examples and preventing leakage along the held-out axis. We hold out 10 gestures for the gesture split and 6 participants for the user split, with an overall train-to-validation/test ratio of approximately 7:3. Figure 3(b) illustrates the split design.

The released dataset includes synchronized EMG and IMU signals, egocentric RGB frames, external RGB-D frames, MANO-based and UmeTrack-based hand-pose annotations, wrist articulation angles, timestamps, and calibration metadata. The prediction target is a 22-DoF joint-angle vector per hand: 20 finger angles and 2 wrist articulation angles. The same target definition and semantic ordering are used for both left and right hands.

4 Benchmark and Baselines

4.1 Benchmark Tasks

EgoEmg defines three prediction tasks under the same 22-DoF target definition and evaluation split axes. The target consists of 20 finger joint angles and 2 wrist articulation angles per hand. The three tasks are: *EMG-to-pose*, which predicts joint angles from EMG windows; *vision-to-pose*, which predicts joint angles from egocentric RGB hand crops; and *EMG+vision fusion*, which uses both modalities to predict the same output. The external RGB-D and IMU streams are included in the release to support future multi-view, inertial, and bimanual modeling, but are not benchmarked in this paper.

4.2 Evaluation Protocol

We evaluate all tasks on the gesture, user, and both splits defined in §3. The primary metric is mean absolute error (MAE) over joint angles:

$$\mathrm{MAE} = \frac{1}{JT} \sum_{j=1}^{J} \sum_{t=1}^{T} |\hat{\theta}_{j,t} - \theta_{j,t}| \quad [^\circ], \tag{1}$$

where J is the number of predicted joint angles and T is the number of evaluated time steps. For wrist analysis, we additionally report MAE on flexion/extension and radial/ulnar deviation. Unless otherwise specified, all errors are reported on the held-out test split. EMG evaluation uses the full predicted trajectory over each EMG window, whereas the vision-only and fusion models are evaluated on the window center frame, where visual evidence is available; cross-modality comparisons should therefore be interpreted with this difference in supervision granularity in mind.

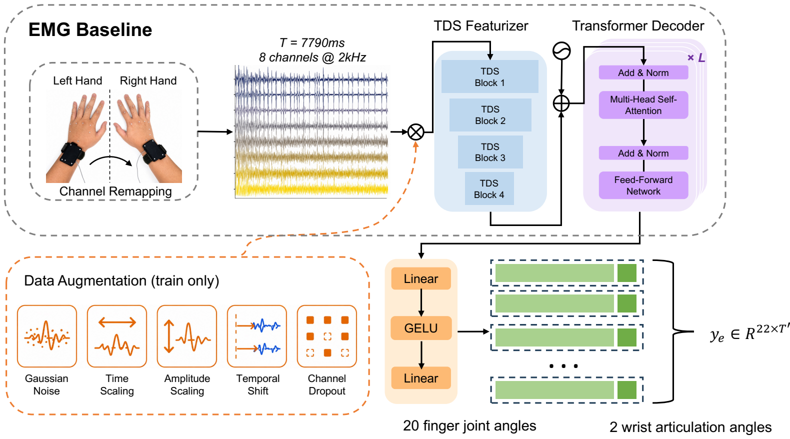

Figure 4: EMGFormer architecture for EMG-to-pose. Bilateral EMG is encoded by a TDS-style temporal convolutional frontend, decoded by a Transformer with RoPE positional encoding, and projected to joint angles including wrist articulation.

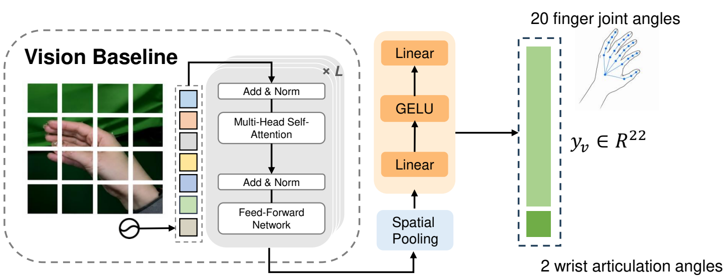

Figure 5: Vision-to-pose baseline. A 256×256 hand crop from the center frame is processed by a ResNet or ViT backbone, followed by an MLP head that regresses 22 joint angles.

4.3 EMG Baseline

As the EMG-to-pose baseline, we introduce EMGFormer (Figure 4), a temporal model that maps raw EMG windows to per-frame joint angles. A TDS-style temporal convolutional frontend [Hannun et al., 2019] downsamples 2 kHz EMG signals to latent features at approximately 37 Hz, followed by a Transformer decoder with rotary positional encoding (RoPE) [Su et al., 2024] to model long-range temporal dependencies. Left-hand EMG is mirrored to the right-hand convention during training. Each forward pass predicts one hand's joint-angle trajectory, including wrist articulation for *EgoEmg*. We provide three model sizes; implementation details, architecture specifications, and ablations are provided in Appendix C.1.

4.4 Vision Baseline

For vision-to-pose, we evaluate seven single-frame reference baselines. Six are generic image backbones across two families: ResNet ({18, 50, 152}) and ViT ({Small, Base, Large}). ResNet backbones are initialized from ImageNet-pretrained weights, while ViT backbones are initialized from DINOv2-pretrained weights. Each generic model takes a 256×256 egocentric hand crop from the window center frame, extracts a feature vector with the backbone, and regresses the 22-DoF target through an MLP prediction head. We additionally evaluate WiLoR [Potamias et al., 2024b] as a hand-specialized vision baseline initialized from a pretrained MANO-based hand-pose checkpoint and fine-tuned on the same center-frame protocol. This separates generic visual reference points from a task-relevant hand-pose estimator while keeping the evaluation target fixed. Model sizes and training configurations are provided in Appendix C.2.

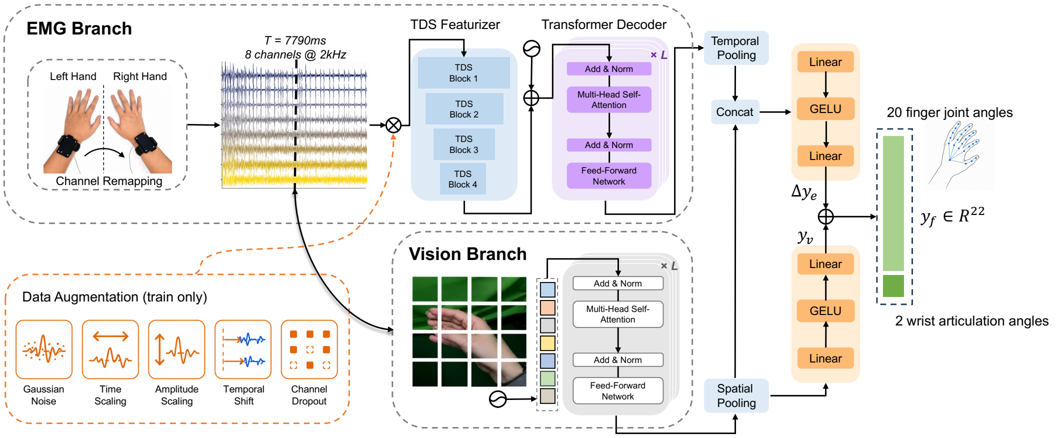

Figure 6: EMG+vision fusion baseline. The vision branch predicts baseline joint angles $\mathbf{y}_v$, while the EMG branch predicts a residual correction $\Delta\mathbf{y}_{\text{emg}}$ from the full EMG window. The residual head is zero-initialized so that training starts from the vision-only baseline.

4.5 Fusion Baseline

The EMG+vision baseline is designed as a simple residual fusion model (Figure 6). The vision branch first predicts a vision-only pose $\mathbf{y}_v$, while the EMG branch predicts an additive correction $\Delta\mathbf{y}_{\text{emg}}$. The final prediction is $\mathbf{y} = \mathbf{y}_v + \Delta\mathbf{y}_{\text{emg}}$. This design makes the fusion model start from a strong visual baseline and learn only the pose information contributed by EMG.

The vision branch extracts a 256-d feature from the center-frame hand crop and uses a prediction head to produce $\mathbf{y}_v \in \mathbb{R}^{22}$. The EMG branch uses EMGFormer to encode the full EMG window into a temporal feature sequence, followed by learned temporal attention pooling to obtain a 256-d EMG feature. The two features are concatenated and passed to a fusion head that predicts $\Delta\mathbf{y}_{\text{emg}} \in \mathbb{R}^{22}$. The last layer of the residual head is zero-initialized so that the model begins from the vision-only prediction. We train the fusion model under a center-supervised protocol: only the window center frame is supervised, matching the vision-only target and enabling direct comparison.

5 Experiments

5.1 EMG Baseline Results on EMG2Pose

EMGFormer also establishes a strong baseline on the EMG2Pose dataset. On the hardest user+stage split, EMGFormer-S achieves $12.34° \pm 1.07°$ MAE, a 22% improvement over the vEMG2Pose baseline [Salter et al., 2024] ($15.8° \pm 1.4°$) with only 55% of the model size. Full results are in Appendix D.3.

5.2 EMG Baseline Results

Table 2 reports EMG-to-pose results on *EgoEmg* for EMGFormer variants and two traditional architectures across three test conditions (gesture/user/both). The baselines are: vEMG2Pose [Salter et al., 2024], a recurrent EMG-to-pose model with an LSTM temporal core; and NeuroPose, a convolutional encoder-decoder baseline [Liu et al., 2021].

EMGFormer variants achieve the best overall accuracy, with EMGFormer-M and EMGFormer-L both outperforming the strongest traditional baseline. The gains are especially pronounced on the gesture split, indicating that EMGFormer better captures gesture-level articulation under the seen-user, held-out-gesture setting. Cross-user generalization remains the central challenge: larger temporal models improve gesture-split performance but do not by themselves close the gap on unseen participants. EMGFormer-M offers the best accuracy–capacity trade-off, matching EMGFormer-L in average MAE with fewer than half the parameters.

Table 2: EMG-to-pose results on *EgoEmg*. MAE is reported in degrees; lower is better. Gesture/User/Both columns report per-user mean ± standard deviation (left+right pooled). Shaded rows highlight EMGFormer variants; bold indicates best per column and underline indicates second best.

Method	Params	Gesture	User	Both	Avg.
EMGFormer variants					
EMGFormer-S	3.5M	$12.8_{\pm 1.4}$	$\mathbf{15.6_{\pm 2.5}}$	$17.4_{\pm 1.2}$	$\underline{14.7}$
EMGFormer-M	6.6M	$\underline{11.8_{\pm 1.6}}$	$\mathbf{15.6_{\pm 1.4}}$	$17.4_{\pm 0.9}$	$\mathbf{14.2}$
EMGFormer-L	16.3M	$\mathbf{11.7_{\pm 1.5}}$	$15.7_{\pm 3.0}$	$17.7_{\pm 1.1}$	$\mathbf{14.2}$
Traditional architectures					
vEMG2Pose	6.0M	$15.0_{\pm 1.4}$	$16.3_{\pm 1.7}$	$\underline{17.3_{\pm 1.3}}$	15.9
NeuroPose	6.4M	$15.8_{\pm 1.2}$	$\underline{15.7_{\pm 1.3}}$	$\mathbf{16.3_{\pm 0.7}}$	16.1

5.3 Vision-to-Pose Results

Table 3 reports vision-only and EMG+vision results on *EgoEmg*. We first analyze the vision-only rows, which include six generic single-frame baselines spanning two backbone families, ResNet and ViT, together with one hand-specialized baseline, WiLoR. Among the generic backbones, ResNet-152 achieves the best vision-only performance across all splits. ResNet-50 also outperforms ViT-Large despite using substantially fewer parameters, suggesting that convolutional backbones remain highly competitive as generic egocentric references in this data regime. Scaling reduces error within both generic backbone families, while cross-user and both splits remain harder than the gesture split across all vision-only models. The relative underperformance of generic ViT regressors may reflect the scale of the available supervised training data, as transformer-based vision models often benefit from larger datasets. In contrast, the hand-specialized WiLoR baseline achieves the strongest standalone visual accuracy, reducing the average MAE to 4.9°. This indicates that task-specific hand-pose pretraining and architecture remain important even in the controlled EgoEmg setting.

5.4 EMG+Vision Fusion Results

We evaluate EMG+vision fusion using the residual architecture described in §4.5, with EMGFormer-S as the EMG backbone. The fusion model predicts the window center frame, where the egocentric RGB observation is available, making its MAE directly comparable to the vision-only baseline under the same supervision target. Table 3 reports matched vision-only and EMG+vision comparisons.

Fusion consistently improves over matched vision-only baselines across all backbone families and splits. The improvement is largest on the gesture split, while user-split gains are smaller, suggesting that EMG is particularly informative for fine articulation but remains affected by inter-subject variability. At the same time, V-WiLoR remains stronger than the current lightweight fusion baselines, indicating that the next benchmark milestone is to test whether EMG also improves a strong hand-specialized visual model rather than only generic backbones. A per-gesture analysis across all 60 gesture classes (Appendix D.5) shows that fusion improves over vision-only on 55 of 60 gestures. Figure 7 illustrates this complementarity qualitatively: when visual evidence is ambiguous, EMG retains pose cues independent of image appearance, and the fused model recovers articulation closer to the ground truth.

For reference under the same center-frame supervision target used by fusion, Appendix D.2 reports additional EMG-only center-frame results for all three emg-only methods sizes; these slightly change the size ranking, with EMGFormer-S becoming the strongest average center-frame model while the three variants remain closely clustered overall.

6 Discussion and Conclusion

EgoEmg provides a multimodal benchmark for hand pose estimation with synchronized bilateral EMG, IMU, egocentric RGB, external RGB-D, and hand-pose annotations including wrist articulation. By using a shared 22-DoF prediction target and unified split axes, the benchmark aligns EMG-to-pose,

Table 3: Vision-only and EMG+vision fusion results on *EgoEmg*. MAE is reported in degrees; lower is better. Subscripts denote standard deviation across users. Avg is the per-sample overall MAE across all test splits. Shaded rows highlight EMG+vision fusion baselines. Bold indicates the better result within each matched vision-only/fusion pair. Underlines mark the strongest generic vision-only backbone, while WiLoR is reported as a hand-specialized vision baseline.

Input	Method	Params	Gesture	User	Both	Avg
ResNet-based comparison						
Vision	V-RN18	11.5M	$5.1_{\pm1.3}$	$6.8_{\pm1.7}$	$6.8_{\pm0.9}$	5.9
EMG+Vision	F-RN18+S	15.5M	$\mathbf{4.6}_{\pm1.3}$	$\mathbf{6.4}_{\pm1.4}$	$\mathbf{6.4}_{\pm0.8}$	**5.4**
ViT-based comparison						
Vision	V-ViT-S	21.9M	$5.0_{\pm1.3}$	$7.3_{\pm2.4}$	$7.2_{\pm0.5}$	6.0
EMG+Vision	F-ViT-S+S	25.9M	$\mathbf{4.5}_{\pm1.3}$	$\mathbf{6.8}_{\pm1.2}$	$\mathbf{6.8}_{\pm0.6}$	**5.5**
Additional vision-only backbones						
Vision	V-RN50	23.5M	$4.5_{\pm1.3}$	$6.2_{\pm2.2}$	$6.2_{\pm0.7}$	5.3
Vision	V-RN152	58.2M	$\underline{4.3}_{\pm1.2}$	$6.1_{\pm2.3}$	$6.1_{\pm0.7}$	5.1
Vision	V-ViT-B	86.2M	$4.8_{\pm1.3}$	$7.0_{\pm2.3}$	$6.9_{\pm0.6}$	5.8
Vision	V-ViT-L	303.8M	$4.4_{\pm1.3}$	$6.6_{\pm2.2}$	$6.6_{\pm0.7}$	5.4
Hand-specialized vision baseline						
Vision	V-WiLoR	631.6M	$\mathbf{3.9}_{\pm1.3}$	$\mathbf{5.6}_{\pm2.4}$	$\mathbf{5.7}_{\pm0.7}$	**4.7**

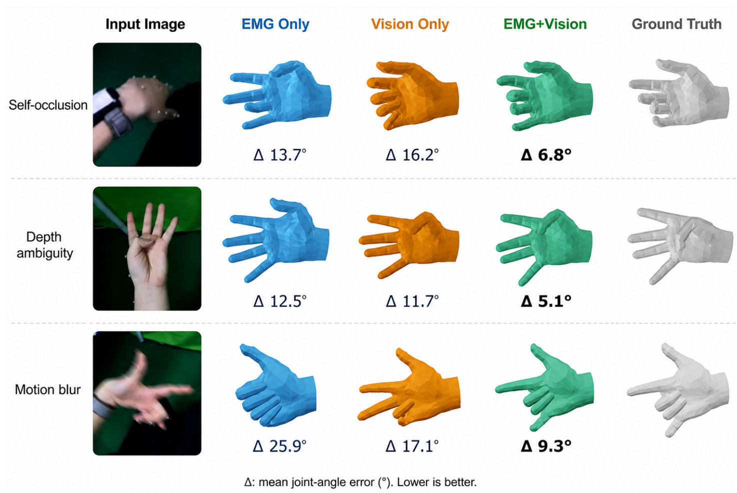

Figure 7: Qualitative comparison of vision-only, EMG-only, and fused predictions on *EgoEmg*. Rows show different test scenarios where visual evidence is degraded (motion blur, self-occlusion, Depth ambiguity). The fused model (green) recovers hand articulation closer to the ground truth, demonstrating the complementary strengths of EMG and vision.

vision-to-pose, and EMG+vision fusion under a common label space while retaining modality-specific supervision protocols.

Our experiments highlight three findings. First, cross-user generalization remains the central challenge for EMG-based pose estimation: errors increase more on unseen participants than on unseen gestures. Second, EMG and vision offer complementary information. Vision provides the stronger standalone signal in the current benchmark, and the hand-specialized WiLoR baseline is the strongest vision-only model, whereas EMG captures pose-relevant cues for fine articulation and improves performance when fused with matched lightweight vision baselines. Third, scaling behavior depends on the data regime. The proposed EMGFormer architecture improves over vEMG2Pose on the original EMG2Pose benchmark, but larger variants do not consistently improve on *EgoEmg*. This suggests that increasing model capacity alone is insufficient under limited subject coverage and motivates large-scale EMG pretraining, cross-dataset transfer, and subject-robust adaptation.

Limitations and future work. The present study has several limitations. First, *EgoEmg* includes 41 participants, which is substantial for synchronized multimodal capture but still limited for modeling the full extent of inter-subject EMG variability. Expanding the dataset to more diverse participants would enable a more systematic study of cross-user robustness. Second, the current vision benchmark is limited to single-frame models. Since *EgoEmg* provides synchronized multi-view video and RGB-D streams, it can support future work on temporal, multi-view, and RGB-D based methods. It also does not yet include rich hand-object interaction with instrumented or densely annotated objects, so future extensions should test whether the same multimodal signals remain helpful when object contact introduces additional occlusion and pose ambiguity. Third, our fusion baselines are intentionally simple and are only evaluated with lightweight generic visual branches. A stronger next step is to test whether EMG also improves specialized vision models such as WiLoR. More expressive architectures, including cross-attention, uncertainty-aware fusion, and explicit temporal alignment, may better exploit the complementary strengths of EMG and vision.

More broadly, *EgoEmg* provides a testbed for studying when physiological signals add information beyond visual observations, especially under unreliable visual evidence such as self-occlusion and fast motion. We hope it will support future work on subject-robust, multimodal hand-pose estimation under a unified benchmark.

Ethics Statement

All participants provided informed consent under an IRB-approved protocol. Participants were compensated at a standard hourly rate. The dataset contains no personally identifiable information: facial regions in egocentric and external camera frames are automatically blurred, and participant identifiers are replaced with anonymous numeric IDs. EMG and motion capture data do not reveal sensitive health information beyond gross motor function. The dataset will be released under a research-use license that prohibits re-identification attempts.

Reproducibility Statement

The repository contains training code, evaluation scripts, and EMGFormer baseline experiments with exact configuration files for all reported results. Appendix C.1 provides per-variant hyperparameters, optimizer settings, and compute requirements. The public release will include dataset documentation, split definitions, preprocessing scripts, benchmark configurations, and an anonymized download link. All EMG2Pose baseline experiments were run on 3 NVIDIA RTX 4090 GPUs with bf16 mixed precision; each EMGFormer variant requires approximately 2–8 GPU-hours for 200 epochs of training. Total compute for all reported experiments is approximately 128 GPU-hours (120 for main baselines + 8 for additional analyses).

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A Dataset details

A.1 Gesture vocabulary

Table 4 lists the complete gesture vocabulary of *EgoEmg* with descriptions.

A.2 MANO shape diversity statistics

Table 5 reports per-dimension statistics of the MANO shape parameters ($\boldsymbol{\beta} \in \mathbb{R}^{10}$) across *EgoEmg*'s 41 participants (82 hands).

A.3 Dataset card and release format

License. The dataset will be released under CC-BY-NC 4.0 for research use, and the baseline code will be distributed under the MIT License. Third-party assets, including the MANO model [Romero et al., 2017] and the training data for *markers2mano*, remain subject to their original licenses.

File schema. Data is organized by episode. Each episode is stored as a parquet file containing synchronized multimodal time series: bilateral EMG (16 channels at 2 kHz, raw and filtered), per-hand IMU, an egocentric RGB frame from a head-mounted wide-angle GoPro (1280×720 at 60 fps), an external RGB-D frame from a ZED 2i sensor (RGB + depth, 1280×720 at 30 fps), MANO pose parameters $\boldsymbol{\theta} \in \mathbb{R}^{48}$ and shape parameters $\boldsymbol{\beta} \in \mathbb{R}^{10}$, wrist flexion/extension and radial/ulnar deviation angles, timestamps, calibration metadata, and split assignments. Video frames are stored as MP4 files and referenced by frame index; each frame includes synchronization diagnostics (`stale` flag and `delta_ms`). Pre-cropped 256×256 left- and right-hand patches, generated by reprojecting mocap keypoints onto the egocentric view and extracting per-hand bounding boxes, are provided as per-episode LMDB archives to support vision-based training without requiring users to re-run the projection pipeline. Windows of 7790 EMG samples are extracted on-the-fly during training.

Access. The dataset will be hosted with a download script and checksum verification. During review, an anonymized access link will be provided.

A.4 EMG Signal Processing

All EMG signals are recorded at 2 kHz. The release includes both raw and filtered EMG for each channel. Filtering is applied per channel in the frequency domain via FFT mask with the following pipeline:

1. Per-channel mean subtraction (DC removal).
2. Narrow-band notch filters at 50 Hz (power line frequency) and 100 Hz (first harmonic).
3. Broadband bandpass filter 20–850 Hz.
4. Soft roll-off above 900 Hz to suppress high-frequency noise.

No amplitude normalization is applied; filtered EMG values are retained in mV. Both raw (`observation.emg.left/right`) and filtered (`observation.emg.left_filtered/right_filtered`) streams are preserved in the release. Training uses the filtered EMG by default.

B Label reconstruction and quality

B.1 Markers2MANO pipeline

This section details the pipeline used to recover MANO parameters from optical motion capture marker positions during *EgoEmg* data collection.

B.1.1 Marker Selection

We select 21 vertices from the MANO mesh surface as virtual markers, distributed across the hand topology to capture wrist, finger, and palm geometry: indices 191, 88, 253, 708, 729 (wrist/palm), 144, 87, 295, 319 (thumb), 220, 365, 407, 445 (index/middle/ring), and 183, 477, 518, 556, 83, 589, 635, 673 (pinky/palm). These indices are consistent across all MANO shape instances $\boldsymbol{\beta}$, enabling direct correspondence between captured 3D marker positions and the canonical MANO mesh.

B.1.2 Graph Transformer Architecture

The primary inference model is a Graph Transformer that maps the 21 marker positions $\mathbf{M} \in \mathbb{R}^{21\times 3}$ to MANO pose $\boldsymbol{\theta} \in \mathbb{R}^{48}$ and shape $\boldsymbol{\beta} \in \mathbb{R}^{10}$ parameters. The architecture consists of:

- **Input embedding**: Each marker position is projected to a latent dimension d via a linear layer ($\mathbb{R}^3 \to \mathbb{R}^d$).
- **Graph positional encoding**: Graph Laplacian eigenvectors ($k = 8$) of the MANO hand-skeleton connectivity graph are projected through a linear layer ($\mathbb{R}^8 \to \mathbb{R}^d$) and *added* to the per-marker input embeddings as positional encoding. This provides each marker with a geometric signature of its location in the hand topology.
- **Structural attention bias**: The shortest-path distances between marker pairs on the hand skeleton graph are embedded and reshaped into a per-head attention bias matrix. This additive bias encourages anatomically adjacent markers (e.g., successive joints along a finger) to attend more strongly to each other.
- **Transformer encoder**: L standard Transformer layers with multi-head self-attention (with the SPD bias added to the pre-softmax attention scores), LayerNorm, and a feed-forward network with $4\times$ expansion ratio ($d \to 4d \to d$). Residual connections wrap each sub-layer.
- **Output heads**: After a final LayerNorm, two separate MLP heads produce the predictions. The *shape head* applies global mean pooling over the 21 marker features followed by a 3-layer MLP to regress $\boldsymbol{\beta} \in \mathbb{R}^{10}$. The *pose head* flattens all per-marker features ($21 \cdot d$) and passes them through a 3-layer MLP to regress 6D rotation representations [Zhou et al., 2019] for the 16 hand joints, which are then converted to axis-angle parameters $\boldsymbol{\theta} \in \mathbb{R}^{48}$.
- **MANO forward pass**: The predicted $\boldsymbol{\theta}$ and $\boldsymbol{\beta}$ are passed through the MANO layer to obtain the final mesh vertices $\mathbf{V} \in \mathbb{R}^{778\times 3}$.

B.1.3 Training and Data Augmentation

The model is trained on hand meshes derived from 11 public datasets—GigaHand [Huang et al., 2024], HOT3D [Banerjee et al., 2024], ARCTIC [Fan et al., 2023], HOI4D [Liu et al., 2022], HanCo [Zimmermann et al., 2021], DexYCB [Yu et al., 2022], H2O [Kwon et al., 2021], InterHand2.6M [Moon et al., 2023], FreiHand [Zimmermann et al., 2019], HO3Dv3 [Hampali et al., 2020], and HO3Dv2 [Hampali et al., 2020]—whose combined raw footage exceeds 195 million frames. To balance the training distribution, we assign each dataset a sampling ratio based on its scale tier: large-scale datasets (100M+ frames, e.g., GigaHand) at 0.14%; medium-scale datasets (1M+, e.g., HOT3D, InterHand2.6M) at 1.5%; and small-scale datasets (100K+, e.g., FreiHand, HO3D) at 15–20%. This yields approximately 0.7M training meshes (Figure 8). This multi-source setup covers a diverse range of hand shapes, poses, and object interactions, ensuring generalization to the gesture space of *EgoEmg*.

The loss function combines two terms evaluated on the MANO mesh output:

$$\mathcal{L} = \lambda_{\text{verts}} \frac{1}{V} \sum_{v=1}^{V} \|\mathbf{v}_v^{\text{pred}} - \mathbf{v}_v^{\text{gt}}\|_2^2 + \lambda_{\text{markers}} \frac{1}{N} \sum_{n=1}^{N} \|\mathbf{m}_n^{\text{pred}} - \mathbf{m}_n^{\text{aug}}\|_2^2, \tag{2}$$

where $\mathbf{v}^{\text{pred}}, \mathbf{v}^{\text{gt}}$ are the predicted and ground-truth MANO mesh vertices ($V = 778$), and $\mathbf{m}^{\text{pred}}, \mathbf{m}^{\text{aug}}$ are the subset of 21 predicted mesh vertices corresponding to the virtual markers and the augmented input markers ($N = 21$), respectively. We set $\lambda_{\text{verts}} = 1.0$ and $\lambda_{\text{markers}} = 5.0$, placing stronger emphasis on accurate marker-level reconstruction.

Data augmentation. To ensure robustness against realistic motion-capture noise, we apply a multi-step augmentation pipeline to the input marker positions prior to model ingestion. The pipeline comprises eight perturbation types organized into three categories; 50% of the samples in each batch bypass augmentation entirely, thereby retaining an unperturbed signal.

- **Structural**: (1) Bone-length perturbation—individual bone segments are randomly scaled by up to $\pm 5\%$ of their length, simulating soft-tissue artifacts. (2) Random global scaling—after

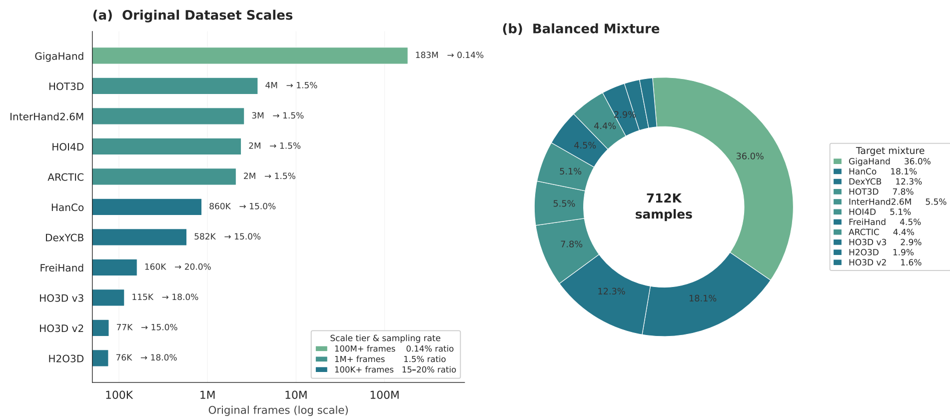

Figure 8: Composition of the *markers2mano* training data. Left: original frame counts of 11 public datasets (log scale), spanning 76K to 183M frames. Each dataset is assigned a sampling ratio by scale tier—0.14% (GigaHand, 100M+), 1.5% (1M+), or 15–20% (100K+)—shown as annotations. Right: target distribution of the training mixture after applying the tiered sampling ratios, yielding approximately 0.7M training meshes.

root-centering, all coordinates are multiplied by a factor in $[0.6, 1.4]$, varying the apparent hand scale.

- **Identity**: (3) Marker swap—spatially proximate markers (within 15 mm) are randomly exchanged with probability 0.3 (capped at three swaps per frame). (4) Marker dropout—up to three markers are dropped and replaced with positions interpolated from their neighbors. (5) Neighborhood blending—each marker is blended with a random convex combination of its neighboring mesh vertices, retaining a self-weight of 0.5.
- **Noise**: (6) Gaussian noise—i.i.d. $\mathcal{N}(0, 1\,\text{mm}^2)$ noise is added to each coordinate, and per-marker dropout is applied with 10% probability. (7) Marker drift—up to three markers receive a systematic spatial offset of up to 5 mm in a random direction. (8) Marker spike—a single marker is randomly displaced to an anomalously distant position (2–5$\times$ the hand scale) with 10% probability, simulating gross tracking failures.

To assess the fidelity of markers2mano reconstruction, we report the reconstruction error of the original markers and virtual vertices of the reconstructed meshes. *markers2mano* shows 3.2mm of per-vertex error on the public dataset, and 4.3mm of per-marker error on the *EgoEmg*. We also manually reproject meshes on the egocentric RGB frames randomly to validate the label quality, as shown in Figure 9.

B.2 Wrist articulation angle computation

Wrist articulation angles are computed directly from motion-capture markers, independent of the MANO reconstruction pipeline. Markers attached to each EMG armband (4 markers on the right wrist, 5 on the left) define a forearm coordinate frame, from which we compute two wrist degrees of freedom: flexion/extension θ_{fe} and radial/ulnar deviation θ_{ru}.

Three non-collinear armband markers define a forearm frame with basis vectors $\mathbf{F}$ (forearm direction) and $\mathbf{L}$ (leftward across dorsal surface), with normal $\hat{\mathbf{N}} = \text{normalize}(\mathbf{F} \times \mathbf{L})$. Let $\hat{\mathbf{H}}$ be the unit vector from wrist to middle finger MCP, and $\hat{\mathbf{H}}_{\text{proj}}$ its projection onto the $(\mathbf{F}, \mathbf{L})$ plane. The wrist angles are:

$$\theta_{\text{fe}} = \arcsin(\hat{\mathbf{H}} \cdot \hat{\mathbf{N}}), \qquad \theta_{\text{ru}} = \text{atan2}\big(\hat{\mathbf{H}}_{\text{proj}} \cdot \hat{\mathbf{L}},\ \hat{\mathbf{H}}_{\text{proj}} \cdot \hat{\mathbf{F}}\big),$$

with extension and radial deviation defined as positive. For the left hand, $\mathbf{N} = \mathbf{L} \times \mathbf{F}$ to maintain consistent sign conventions.

B.3 Joint angle conversion from MANO to UmeTrack / EMG2Pose format

In addition to MANO parameters $(\boldsymbol{\theta}, \boldsymbol{\beta}, \mathbf{t})$, *EgoEmg* provides per-frame joint angles in the 20-DoF scalar format used by EMG2Pose and UmeTrack [Salter et al., 2024, Han et al., 2022], enabling direct comparison with prior EMG-to-pose benchmarks that use scalar joint angle representations. The conversion is formulated as landmark-level inverse kinematics.

Conversion pipeline. Given MANO pose and shape parameters for a frame, we first run MANO forward kinematics to obtain 21 joint positions. We select 20 of these joints (all except the thumb CMC, which has no direct UmeTrack landmark counterpart) and align them into the UmeTrack coordinate frame via a Procrustes transformation precomputed from the two models' rest poses. We then optimize 20 scalar joint angles $\mathbf{a} \in \mathbb{R}^{20}$ such that the UmeTrack forward kinematics produces landmark positions matching the aligned MANO targets:

$$\mathcal{L}(\mathbf{a}) = \frac{1}{20} \sum_{i=1}^{20} \left\| \mathbf{p}_i^{\text{umetrack}}(\mathbf{a}) - \mathbf{p}_i^{\text{mano}} \right\|_2^2, \tag{3}$$

where $\mathbf{p}_i^{\text{mano}}$ are the Procrustes-aligned MANO joint positions and $\mathbf{p}_i^{\text{umetrack}}(\mathbf{a})$ are the corresponding UmeTrack landmarks computed via the UmeTrack forward kinematics (scalar joint angles $\rightarrow$ axis-angle rotations via Rodrigues $\rightarrow$ skinning).

Optimization. We minimize $\mathcal{L}(\mathbf{a})$ with LBFGS (strong Wolfe line search, learning rate 0.05, maximum 30 iterations). Per-joint range constraints from the UmeTrack hand model are enforced via hard clamping of $\mathbf{a}$ at each iteration. The optimization runs per-episode on GPU with batch size 256; episodes exceeding GPU memory are processed in smaller chunks.

Output. The resulting 22-DoF prediction target combines 20 finger joint angles (indices 0–19) and 2 wrist articulation angles (indices 20–21). Table 6 lists the complete semantic ordering per dimension. The conversion script is included in the dataset release.

We show reconstruction samples in Figure 9.

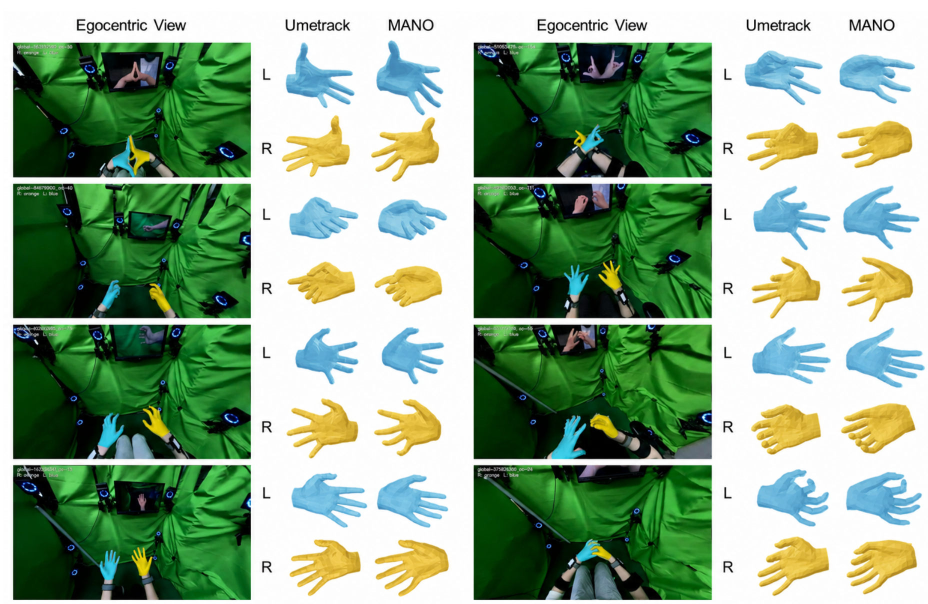

Figure 9: Mesh reprojection and visualization of UmeTrack and MANO mesh reconstruction on *EgoEmg*. The reconstructed MANO meshes (from *markers2mano*) are overlaid on egocentric RGB frames, demonstrating accurate hand pose reconstruction and realistic mesh-to-image alignment.

C Training and implementation details

C.1 EMGFormer training details

Optimization. All EMGFormer variants are trained for 200 epochs with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4}), initial learning rate 10^{-4} with cosine annealing to 5×10^{-6}, batch size 800, and bf16 mixed precision. We use a linear warmup for the first 500 steps. All experiments were conducted on six NVIDIA RTX 4090 GPUs (24 GB each).

Data augmentation. Applied on-the-fly during training: channel dropout (25% probability per channel), frequency masking (3 masks of up to 128 bins), additive Gaussian noise (SNR 25–35 dB with 50% probability), and temporal jitter (± 40 ms).

Architecture and per-size configuration. Table 7 summarizes the architecture and training configuration per EMGFormer variant.

TDS featurizer. The TDS-style temporal convolutional frontend [Hannun et al., 2019] maps the raw 2 kHz EMG window ($T = 7790$ samples, 16 channels) to a 256-dimensional feature sequence. The architecture consists of two initial strided Conv1d blocks followed by two TDS stages. Each TDS stage begins with a strided Conv1d for temporal subsampling, followed by a TDSConvEncoder composed of alternating depth-separable temporal convolutions (reshaping the channel axis into a 2D feature map) and channel-wise fully-connected blocks. Squeeze-and-excitation is applied globally after each TDS stage. The layer-wise configuration is:

1. Conv1d: $16 \rightarrow 256$, kernel 11, stride 5
2. Conv1d: $256 \rightarrow 256$, kernel 5, stride 2
3. TDS Stage 1: in-conv kernel 9, stride 5; 1 block of TDSConv (kernel width 5, channels 8, feature width 32)
4. TDS Stage 2: in-conv kernel 3, stride 1; 1 block of TDSConv (kernel width 3, channels 8, feature width 32)

The final output is a $(256, 146)$ feature map, corresponding to an effective frame rate of approximately 37 Hz with a receptive field spanning the full input window.

C.2 Vision and fusion training details

Vision baseline. We provide six fully fine-tuned generic vision backbones: ResNet-$\{18, 50, 152\}$ and ViT-$\{$Small, Base, Large$\}$. ResNet backbones are initialized from ImageNet-pretrained weights, while ViT backbones are initialized from DINOv2-pretrained weights. Each generic backbone extracts a feature vector from the window center frame crop (256×256), which is fed through a prediction head to regress 22 joint angles. The prediction head is a 2-layer MLP: Linear($d_{\text{backbone}} \rightarrow 512$) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.1) $\rightarrow$ Linear$(512 \rightarrow 22)$. We additionally fine-tune WiLoR [Potamias et al., 2024b] as a hand-specialized baseline using its pretrained MANO-based checkpoint, with the same center-frame supervision target and output dimensionality. Generic backbones use AdamW with cosine annealing and bf16 mixed precision on 6 NVIDIA RTX 4090 GPUs; WiLoR uses AdamW with learning rate 10^{-5} and cosine annealing over 30 epochs. Table 8 lists the per-backbone configuration.

Fusion baseline. The fusion model uses the residual architecture described in §4.5. The vision pathway (backbone, projection layer, and vision head) can be bootstrapped from a vision-only checkpoint, and the EMG pathway (EMGFormer-S, comprising a TDS featurizer and a Transformer decoder) can be initialized from an EMG-only checkpoint. The fusion projection and the residual regression head are randomly initialized, with the last layer of the head set to zero so that the model starts from the vision-only baseline ($\Delta \mathbf{y}_{\text{emg}} \approx 0$). The model predicts only the center frame of the window. Training is performed with AdamW (learning rate 1×10^{-4}, cosine annealing schedule), a batch size ranging from 64 to 600 depending on the backbone, and 200–300 epochs, using bf16 mixed precision on six NVIDIA RTX 4090 GPUs.

D Extended experimental results

D.1 Per-hand breakdown of EMG baseline results

Table 9 reports per-hand MAE for all methods on *EgoEmg*, complementing the aggregate results in Table 2.

Left and right hand MAE agree within 0.6° across all models and splits, confirming that the shared-weight bilateral design and left-to-right mirroring strategy produce symmetric performance. The per-hand consistency supports the validity of processing both hands through a single model.

D.2 Center-frame EMG-only reference

To partially normalize the supervision granularity gap discussed in §4.1, we additionally evaluate EMG-only models on the window center frame rather than over the full predicted trajectory. Table 10 reports these center-frame MAE values using the same gesture, user, and both splits as the main benchmark. The center-frame results are comparable but not identical to the trajectory-level evaluation: they slightly change the ranking among model sizes, with EMGFormer-S achieving the best average MAE and the best user-split result, while EMGFormer-L remains strongest on the gesture and both splits. All three variants remain tightly clustered within 0.4° average MAE.

D.3 EMG2Pose Benchmark Results

Table 11 reports EMGFormer results on the EMG2Pose dataset standard splits.

EMGFormer-L achieves the best user+stage MAE of $12.25° \pm 1.08°$, improving over the vEMG2Pose baseline [Salter et al., 2024] ($15.8° \pm 1.4°$) by 22% while using $2.5\times$ more parameters. CLDM [Verma, 2025] reports a user+stage MAE of $14.7° \pm 1.4°$, and VQ-MyoPose [Lovecchio et al., 2025] was evaluated on a 12-subject subset of EMG2Pose rather than the full 193-participant benchmark. EMGFormer-L reduces the stage split MAE to $9.28° \pm 1.09°$, a 39% improvement over vEMG2Pose, indicating stronger temporal pattern modeling for unseen gesture stages.

D.4 Per-joint error analysis

Table 12 presents the per-finger and per-joint-group mean absolute error (MAE) for EMGFormer variants on *EgoEmg*. Across all three model sizes, distal joints and wrist deviation yield the lowest errors (best MAE $10.0°$ for both), whereas the pinky and mid-phalanx groups are the most challenging (best MAE $16.7°$ and $20.7°$, respectively). This pattern is consistent with hand kinematics and sensing difficulty. Distal joints usually have a more constrained range of motion, making their angles easier to regularize from EMG signals. In contrast, mid-phalanx joints capture a larger portion of finger flexion and therefore exhibit greater motion variability. The higher error on the pinky is also expected: pinky motion is often weaker, more coupled with the ring finger, and less independently represented in wrist-level EMG, making it harder to distinguish from neighboring-finger activation patterns.

For comparison, Table 13 details the corresponding per-joint MAE on the EMG2Pose dataset. Because EMG2Pose lacks wrist articulation labels, our evaluation here is restricted to finger joints.

A cross-dataset comparison shows that mid-phalanx and pinky joints are consistently among the most challenging groups across both benchmarks. However, the easiest groups differ between datasets: proximal joints achieve the lowest MAE on EMG2Pose, whereas distal joints and wrist deviation are easiest on *EgoEmg*. The overall error magnitudes are also higher on *EgoEmg*—ranging from $10.0°$ to $21.6°$, compared with $8.5°$ to $14.2°$ on EMG2Pose—which likely reflects differences in participant count, gesture diversity, sensor setup, and the inclusion of wrist articulation in *EgoEmg*'s 22-DoF prediction target.

D.5 Per-Gesture Modality Complementarity Analysis

To characterize where EMG provides complementary information beyond vision, we evaluate all three modalities—vision-only, EMG-only, and F-RN18+S fusion—on a per-gesture basis across the full test set (all splits, both hands). Following the updated vocabulary in Appendix A.1, we group the 60 gesture classes into three semantic families: single-hand (30 gestures), symmetric bimanual (22), and asymmetric bimanual (8). Table 14 reports sample-weighted per-family averages across all three modalities.

Three patterns emerge. First, the fusion gain is systematic: fusion improves over vision-only on 55 of 60 gesture classes, ruling out the possibility that the aggregate improvement is driven by a few outlier gestures. Second, the largest family-level gain occurs on single-hand gestures ($\Delta = +0.50°$), while asymmetric bimanual gestures are close behind ($\Delta = +0.48°$). This suggests that EMG is most useful when the prediction depends on subtle within-hand articulation or asymmetric cross-hand coordination that is harder to infer from a single RGB crop. Symmetric bimanual gestures show the smallest gain ($\Delta = +0.35°$), as two-hand interactions of this type are often easier to discriminate visually. Third, even for the few gestures where fusion slightly underperforms vision-only (Rock, $-0.22°$; Typing, $-0.21°$; Clap, $-0.12°$), the degradation is negligible and involves large-scale hand silhouettes whose visual features are already highly discriminative. Figure 10 visualizes the per-gesture complementarity: nearly all points lie below the diagonal, confirming that fusion provides a systematic, if modest, correction across the gesture vocabulary.

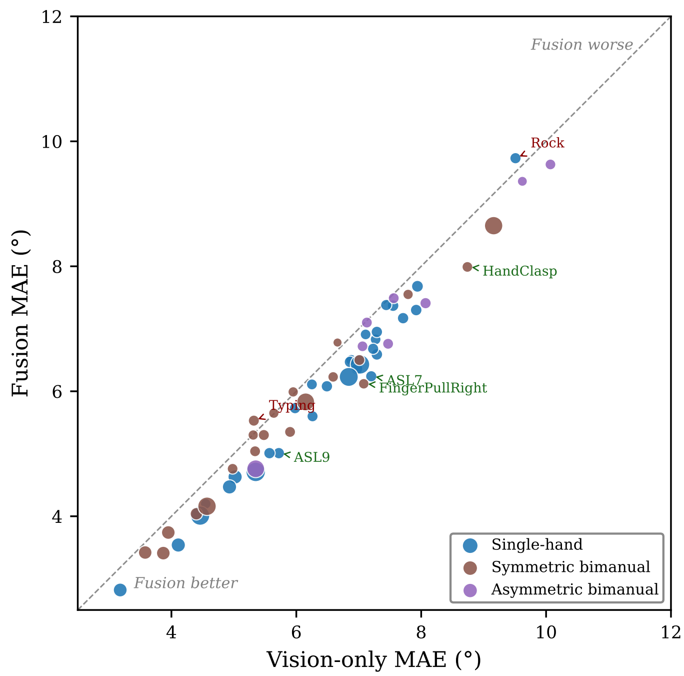

Figure 10: Per-gesture vision-only vs. fusion MAE on *EgoEmg* (F-RN18+S). Each point is one gesture class; point size reflects sample count; color indicates the semantic gesture family. The diagonal marks parity; points below the diagonal indicate fusion improvement. Top-4 most-improved gestures and the 2 worsened gestures are annotated.

Table 15 provides the complete per-gesture breakdown for all 60 gesture classes, grouped by semantic family and sorted by descending fusion gain within each family.

Table 4: Complete gesture vocabulary of *EgoEmg* (60 classes) with per-gesture descriptions.

Gesture	**Description**
Single-hand (30 gestures)	
ASL1–ASL9	American Sign Language digits 1 through 9
Claw3	Three-finger claw pose (index + middle + ring flexed)
Claw5	Five-finger claw pose (all fingers flexed)
FreeAction	Free-form unconstrained hand action
ILY	“I Love You” hand sign (thumb, index, pinky extended)
IndexBow	Index finger flexion/extension bowing
IndexMiddleClaw	Index and middle fingers clawed, others extended
JoystickCircle	Circular joystick manipulation on the palm with thumb
JoystickSlide	Linear joystick sliding on the palm with thumb
MiddleBow	Middle finger flexion/extension bowing
Nine	Hand forming number 9 (four fingers curled, Index up)
PalmYaw	Palm facing up/down rotation about the yaw axis
PinchMiddle	Thumb to middle fingertip pinch
PinkyBow	Pinky finger flexion/extension bowing
Rest	Relaxed hand in neutral resting posture
RingAndThumb	Ring finger touching thumb tip
RingBow	Ring finger flexion/extension bowing
Rock	“Rock on” hand sign (index and pinky extended)
Thumb	Thumb up
nocontact_disperse_palm	Fingers spread apart with palm open, no contact
nocontact_free	Free-form finger motion without contact
nocontact_grab	Simulated grasping motion without without contact
Symmetric bimanual (22 gestures)	
Clap	Both hands clapping together symmetrically
CrossHand	Both hands crossed fingers
CrossStretch	Hands opposite fingers and stretched
FingerTipTouch	Matching fingertips of both hands touching
FistBump	Two fists bumping together
Gaming	Both hands holding a game controller
HandClasp	Hands clasped together
HandRub	Rubbing palms together
IndexTapping	Both index fingers tapping in the air
Kiss	Fingertips of both hands touching(Thumb and Middle), then separating
PalmStack	One palm stacked on top of the other
Prayer	Palms pressed together in prayer position
MiddleOppo	Middle and Ring finger of one hand opposes the other hand
SymOpen	Both hands opening symmetrically from closed to open
SymSwing	Both hands swinging symmetrically side to side
ThumbWrestle	Thumbs wrestling each other
Typing	Both hands typing on a virtual keyboard
raw	Unconstrained bimanual hand movement
Asymmetric bimanual (8 gestures)	
FingerPullLeft	Left hand pulls right-hand fingers
FingerPullRight	Right hand pulls left-hand fingers
PalmRoll	Palms facing, rolling around each other
PinkyHook	Pinky fingers hooked together
Squeeze	Both hands squeezing an imaginary object
Beijing	Two hands crossing to make a "Bei" Chinese character
Checky	Two hands Thumb make "Checky" sign
PairClaw	Both hands clawing with asymmetric finger configurations
PairOK	Both hands forming OK signs
Picture	Hands forming a picture frame rectangle
PinchWring	Two hand pinch and wring
ThumbOppo	Thumbs of both hands in opposition at different orientations

Table 5: Per-dimension MANO shape coefficient statistics across *EgoEmg* participants. MANO shape parameters are PCA-derived and have no fixed semantic interpretation.

Dimension	Mean	Std	Min	Max	P95–P5 range
β_1	1.087	1.051	-0.966	3.034	4.000
β_2	1.505	0.793	0.183	3.506	3.323
β_3	-0.981	1.076	-3.804	2.547	6.351
β_4	-0.537	1.069	-3.077	2.050	5.128
β_5	1.054	0.996	-0.461	4.638	5.099
β_6	-2.269	1.711	-10.253	0.450	10.703
β_7	0.727	1.437	-1.735	8.254	9.989
β_8	0.371	1.000	-1.802	4.683	6.485
β_9	0.365	1.464	-2.978	5.344	8.322
β_{10}	-0.738	0.553	-3.308	0.366	3.674

Table 6: Complete 22-DoF joint angle target semantics. Each dimension is reported in degrees. FE = flexion/extension, AA = abduction/adduction. Indices 0–3 cover the thumb, 4–7 the index finger, 8–11 the middle finger, 12–15 the ring finger, and 16–19 the pinky.

Index	Finger	Joint	Motion
0	Thumb	CMC	FE
1	Thumb	CMC	AA
2	Thumb	MCP	FE
3	Thumb	IP	FE
4	Index	MCP	AA
5	Index	MCP	FE
6	Index	PIP	FE
7	Index	DIP	FE
8	Middle	MCP	AA
9	Middle	MCP	FE
10	Middle	PIP	FE
11	Middle	DIP	FE
12	Ring	MCP	AA
13	Ring	MCP	FE
14	Ring	PIP	FE
15	Ring	DIP	FE
16	Pinky	MCP	AA
17	Pinky	MCP	FE
18	Pinky	PIP	FE
19	Pinky	DIP	FE
20	–	Wrist	Flexion/Extension
21	–	Wrist	Radial/Ulnar deviation

Table 7: Architecture and training configuration per EMGFormer variant. "Params" includes the TDS encoder, Transformer decoder, and regression head.

Variant	Layers	Hidden dim	Heads	FFN dim	Params	GPU-hours
EMGFormer-S	3	256	4	512	3.5M	2
EMGFormer-M	6	256	8	1024	6.6M	4
EMGFormer-L	8	384	12	1536	16.3M	8

Table 8: Vision backbone configuration. All models take 256×256 center-frame crops as input and are evaluated on the same 22-DoF center-frame target. Generic ResNet/ViT models share the same MLP prediction head; WiLoR uses its native hand-pose backbone with a 22-DoF regression head.

Backbone	Params	d_{backbone}	LR	Batch size	Epochs
ResNet-18	11.5M	512	10^{-3}	900	150
ResNet-50	23.5M	2048	10^{-3}	400	150
ResNet-152	58.2M	2048	10^{-3}	180	150
ViT-S/14	21.9M	384	10^{-4}	360	150
ViT-B/14	86.2M	768	10^{-4}	160	150
ViT-L/14	303.8M	1024	10^{-4}	50	150
WiLoR	631.6M	–	10^{-5}	–	30

Table 9: Per-hand EMG-to-pose MAE on *EgoEmg*. MAE is reported in degrees; lower is better. Left and right hands are processed independently with shared model weights, and left-hand EMG is mirrored to the right-hand convention. Rows are grouped by method family; shaded rows highlight EMGFormer variants.

Method	Hand	Gesture	User	Both
EMGFormer variants				
EMGFormer-S	Left	12.4	14.7	16.5
	Right	13.2	16.6	18.4
EMGFormer-M	Left	11.5	15.5	17.0
	Right	12.1	16.6	17.8
EMGFormer-L	Left	11.4	15.3	17.6
	Right	12.0	16.0	17.7
Traditional architectures				
vEMG2Pose	Left	14.7	15.7	17.2
	Right	15.2	17.0	17.5
NeuroPose	Left	15.9	15.4	15.7
	Right	15.7	16.0	16.9

Table 10: EMG-only center-frame reference results on *EgoEmg*. MAE is reported in degrees; lower is better. Columns report per-user mean $\pm$ standard deviation (left+right pooled). Avg is the per-sample overall MAE across all test splits. These runs use the same center-frame target as the vision-only and fusion models, but remain EMG-only. Bold indicates the best mean per column.

Method	Params	Gesture	User	Both	Avg.
EMGFormer-S	3.5M	$12.5_{\pm 1.7}$	$\mathbf{15.5_{\pm 2.5}}$	$17.0_{\pm 1.4}$	**14.5**
EMGFormer-M	6.6M	$12.9_{\pm 1.7}$	$15.7_{\pm 2.6}$	$17.7_{\pm 1.6}$	14.9
EMGFormer-L	16.3M	$\mathbf{12.6_{\pm 1.6}}$	$16.1_{\pm 1.6}$	$\mathbf{17.4_{\pm 1.2}}$	14.6

Table 11: EMG-only baseline results on EMG2Pose test splits. MAE is reported in degrees; lower is better. EMGFormer values report per-user mean $\pm$ standard deviation across users (n=158 for Stage, n=20 for User and User+Stage). †Numbers reported in the original papers. ‡Trained and evaluated on a 12-subject subset of EMG2Pose (12/27 stages); not directly comparable to full-dataset results. Shaded rows highlight EMGFormer variants; bold indicates the best EMGFormer value per column.

Method	Params	User	Stage	User+Stage
Prior work				
SensingDynamics‡	–	$15.5_{\pm 1.4}$	$18.8_{\pm 1.6}$	$18.7_{\pm 1.6}$
NeuroPose‡	6.4M	$13.2_{\pm 1.1}$	$17.2_{\pm 1.7}$	$17.5_{\pm 1.5}$
vemg2pose†	6.4M	$12.2_{\pm 1.3}$	$15.2_{\pm 1.6}$	$15.8_{\pm 1.4}$
CLDM† [Verma, 2025]	–	$11.3_{\pm 1.0}$	$14.3_{\pm 1.50}$	$14.7_{\pm 1.4}$
EMGFormer variants				
EMGFormer-S	3.5M	$12.45_{\pm 1.06}$	$11.10_{\pm 1.19}$	$12.34_{\pm 1.07}$
EMGFormer-M	6.6M	$12.40_{\pm 1.08}$	$10.18_{\pm 1.13}$	$12.39_{\pm 1.08}$
EMGFormer-L	16.3M	$\mathbf{12.26_{\pm 1.08}}$	$\mathbf{9.28_{\pm 1.09}}$	$\mathbf{12.25_{\pm 1.08}}$

Table 12: Per-joint MAE (°) on *EgoEmg*, left and right hands pooled within each split, then per-sample weighted average across gesture, user, and both splits. Lower is better. Bold indicates the best mean per row.

Joint group	EMGFormer-S	EMGFormer-M	EMGFormer-L
Distal	10.2	10.1	**10.0**
Wrist deviation	10.8	10.1	**10.0**
Thumb	12.8	**12.3**	**12.3**
Index	13.4	**13.3**	13.3
Proximal	13.7	**13.3**	13.4
Ring	15.2	14.7	**14.6**
Middle finger	15.2	14.8	**14.8**
Wrist flexion	15.7	14.9	**14.6**
Pinky	17.3	**16.7**	17.0
Mid-phalanx	21.6	**20.7**	**20.7**

Table 13: Per-joint MAE (°) on the EMG2Pose dataset, per-sample weighted average across all three test splits (stage, user, and user+stage). Because EMG2Pose lacks wrist labels, only finger joints are reported. Bold indicates the best mean per row.

Joint group	EMGFormer-S	EMGFormer-M	EMGFormer-L
Proximal	9.3	8.9	**8.5**
Index	10.8	10.3	**9.8**
Thumb	11.1	10.6	**9.9**
Middle finger	11.1	10.5	**10.0**
Ring	11.1	10.6	**10.1**
Distal	13.3	12.8	**12.2**
Pinky	13.5	12.9	**12.3**
Mid-phalanx	14.2	13.4	**12.6**

Table 14: Per-gesture-family modality comparison on *EgoEmg* (F-RN18+S fusion). Values are sample-weighted averages across all test splits; MAE is in degrees (lower is better). $\Delta = \text{MAE}_{\text{vision}} - \text{MAE}_{\text{fusion}}$; positive Δ indicates fusion improvement. Gesture families follow the vocabulary in Appendix A.1.

Gesture family	#G	#Samp.	Vis.	EMG	Fusion	Δ
Single-hand	30	2,406	6.0	15.0	5.5	$\mathbf{+0.50}$
Symmetric bimanual	21	1,432	6.0	14.0	5.6	$+0.35$
Asymmetric bimanual	9	440	6.7	14.9	6.2	$+0.48$
Overall	60	4,278	6.1	14.6	5.6	$+0.45$

Table 15: Full per-gesture modality breakdown on *EgoEmg* using F-RN18+S fusion. MAE is in degrees. $\Delta = \text{MAE}_{\text{vis}} - \text{MAE}_{\text{fus}}$. Gestures are grouped by family and sorted by descending Δ within each family. Gesture descriptions are provided in Appendix A.1.

ID	Name	N	Vis.	EMG	Fus.	Δ
Single-hand gestures						
6	ASL7	36	7.2	16.1	6.2	+0.96
8	ASL9	38	5.7	17.5	5.0	+0.71
12	ILY	40	7.3	19.2	6.6	+0.70
13	IndexBow	34	6.3	20.7	5.6	+0.66
20	PinchMiddle	324	5.3	12.2	4.7	+0.64
26	Thumb	40	7.9	24.8	7.3	+0.62
7	ASL8	42	7.0	14.8	6.4	+0.61
28	nocontact_free	292	6.8	14.2	6.2	+0.61
18	Nine	320	7.0	18.1	6.4	+0.59
17	MiddleBow	92	4.1	11.1	3.5	+0.57
16	JoystickSlide	38	5.6	19.5	5.0	+0.56
23	RingAndThumb	36	7.2	16.5	6.7	+0.55
9	Claw3	38	7.7	17.3	7.2	+0.54
27	nocontact_disperse_palm	92	4.9	12.2	4.5	+0.46
15	JoystickCircle	308	4.5	12.2	4.0	+0.45
2	ASL3	30	7.3	19.6	6.8	+0.44
11	FreeAction	38	6.5	15.8	6.1	+0.41
5	ASL6	40	6.9	18.1	6.5	+0.39
14	IndexMiddleClaw	92	5.0	12.9	4.6	+0.39
1	ASL2	34	6.9	17.7	6.5	+0.38
19	PalmYaw	82	3.2	9.8	2.8	+0.36
4	ASL5	28	4.5	14.6	4.2	+0.35
24	RingBow	38	7.3	16.9	7.0	+0.34
0	ASL1	44	7.9	17.6	7.7	+0.26
10	Claw5	36	6.0	15.6	5.7	+0.25
22	Rest	30	7.1	17.4	6.9	+0.20
21	PinkyBow	40	7.5	16.6	7.4	+0.18
3	ASL4	32	6.2	18.0	6.1	+0.14
29	nocontact_grab	36	7.4	18.6	7.4	+0.06
25	Rock	36	9.5	17.5	9.7	-0.22
Symmetric bimanual gestures						
34	FingerPullRight	28	7.1	13.6	6.1	+0.96
38	HandClasp	30	8.7	16.4	8.0	+0.75
39	HandRub	32	5.9	16.0	5.3	+0.55
36	FistBump	256	9.2	23.7	8.7	+0.51
43	PinkyHook	32	7.0	16.0	6.5	+0.51
44	Prayer	78	3.9	9.5	3.4	+0.46
42	PalmStack	262	4.6	10.2	4.2	+0.41
41	PalmRoll	54	4.4	12.1	4.0	+0.36
47	SymSwing	28	6.6	15.7	6.2	+0.36
45	Squeeze	230	6.2	12.4	5.8	+0.32
40	IndexTapping	32	5.3	13.0	5.0	+0.30
37	Gaming	26	7.8	15.9	7.5	+0.24
46	SymOpen	30	5.0	12.6	4.8	+0.22
51	Kiss	78	4.0	10.4	3.7	+0.21
48	ThumbWrestle	34	5.5	14.6	5.3	+0.18
35	FingerTipTouch	76	3.6	7.5	3.4	+0.16
32	CrossStretch	26	5.3	11.9	5.3	+0.01
31	CrossHand	26	5.6	12.8	5.7	-0.01
33	FingerPullLeft	26	6.0	13.6	6.0	-0.04
30	Clap	14	6.7	14.3	6.8	-0.12
49	Typing	34	5.3	11.9	5.5	-0.21
Asymmetric bimanual gestures						
58	PinchWring	32	7.5	17.8	6.8	+0.71
55	PairClaw	36	8.1	17.4	7.4	+0.66
52	MiddleOppo	226	5.3	12.3	4.8	+0.59
53	Beijing	30	10.1	19.2	9.6	+0.44
59	ThumbOppo	30	7.1	17.5	6.7	+0.34
54	Checky	22	9.6	21.1	9.4	+0.26
56	PairOK	32	7.6	15.6	7.5	+0.07
57	Picture	32	7.1	16.7	7.1	+0.03

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